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Environmental contours for design of ice-capable vessels

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ABSTRACT: The environmental contour method is an established method for design of ships and marine structures. In this method, environmental contours for a given return period are developed in order to identify the environmental conditions which are associated with the most critical structural responses, and then the long-term extreme response for the same return period can be effectively estimated based on the response statistics of the most critical (short-term) responses. In this work, the environmental contour method is described for design of ice-capable vessels in Arctic regions. In particular, for vessels sailing in Arctic regions, the most dangerous condition is associated with ice ridges. Based on the probability distributions of key parameters of the first-year ice ridges which determine the ice ridge loads, the environmental contours are developed. The effects of the correlation between different environmental variables are discussed. A simple case is proposed to illustrate application of the environmental contour method for design of ice-capable vessels.

1 INTRODUCTION

For ships and marine structures subjected to environmental loads, such as wind, wave and ice forces, etc., evaluation of the extreme response over their specific lifetime is necessary and important at the design stage. The full long-term analysis which accounts for the structural response from each short-term environmental condition and the occurrence rate of each short-term condition is recognized as the most reliable and accurate approach. However, such long-term analysis is time consuming, especially for large and complex systems (Leira, 2008).

In order to improve the computational efficiency, the long-term analysis can be simplified either by reducing the computation cost of short-term analysis (Chai et al., 2016) or by developing approximate method that requires a lower number of short-term analysis, such as the IFORM (inverse first order reliability method) (Giske et al., 2017) and the environmental contour method (Haver and Winterstein, 2009).

The environmental contour method offers a simplified and fast way to estimate the long-term extreme response and has been widely used for design of ships and marine structures at the early design stage (Li et al., 2016, DNV, 2010, Baarholm et al., 2010). An environmental contour is the locus of all environmental parameters correspond to a given annual exceedance probability along which extreme responses with the corresponding return period should lie. Traditionally, establishment of such environmental contour for a given return period is based on the IFORM and the joint distribution of the environmental parameters.

The main advantage for the environmental contour method lies in the fact that the structural response is uncoupled from the description of the environmental variables. As a result, the long-term extreme response for a given return period can be approximately estimated on the basis of response statistics associated with the most critical short-term environmental conditions along the environmental contour, which corresponds to a given return period. Therefore, in this method, short-term analyses are performed only for a few conditions on the environmental contour and then numerical simulations or experiments are performed in order to identify the most critical short-term response for further application (Winterstein et al., 1993).

In this work, the concept of environmental contour is applied for design of ice-capable vessels operating in Arctic regions. For ships sailing in sea ice areas without icebergs, ice ridges are assumed to pose a major threat to the vessel since they determine and govern the design loads for the vessel (Høyland, 2014). Key parameters of the ice ridge which determine the main loads of the ship-ice ridge interaction process are identified at first. Then, probabilistic models are applied to describe the distributions of these key parameters in order to develop the environmental contours.

The procedure of extreme response prediction based on the environmental contour method is simply described in this work. Finally, a simple case is given to illustrate the development of a

desired environmental contour used for design of ice-capable vessels.

2 ENVIROMENTAL CONTOUR METHOD

In this section, construction of the environmental contour for a given return period and application of the environmental contour method are presented. Assume that for the N-year return period, the corresponding failure probability of the structure response is given as P_f. As for the traditional IFORM to generate the environmental contour with a N-year return period, a n-dimensional (n is the dimensionality of the environment parameter vector) hypersphere with radius β_F is first created in the U space, with the value of β_F being given as (Haver and Winterstein, 2009):

$$\beta_F = \Phi^{-1}(1 - P_f) \tag{1}$$

where Φ represents the cumulative density function of the standard normal distribution.

Then, the n-dimensional hypersphere is transformed into the physical parameter space by consideration of the joint distribution of the environmental parameters in order to get the environmental contour. Figure 1 presents an example of an environmental contour in the physical parameter space.

After establishment of the environmental contour with the N-year return period, a limited set of design conditions, i.e. short-term environmental conditions, are selected on the environmental contour (e.g. see Figure 1). Time consuming numerical calculations (or expensive experiments) for structural responses are only needed for these selected environmental conditions. The most critical short-term response is identified in order to estimate the long-term extreme response with the N-year return period. The main process of applying the environmental contour method for long-term extreme response approximation is presented in Figure 2.

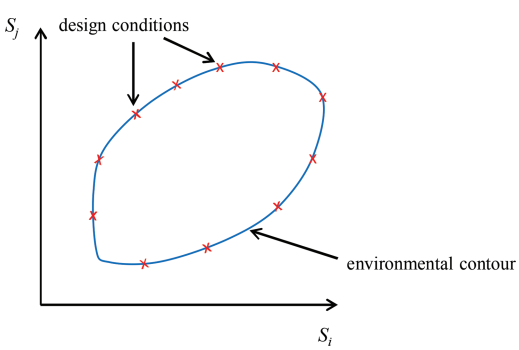

Figure 1. Environmental contour in the physical parameter space and selected environmental design conditions.

3 FIRST-YEAR ICE RIDGE

A ridge is a line or wall of broken ice features forced by pressure or shear. When first formed, an ice ridge is simply a pile of unconsolidated ice blocks. Then, these blocks may become to some extent consolidated by refreezing processes and form the ice ridge. Figure 1 illustrates a typical ice ridge, which consists of two parts: the sail and the keel. The sail part is above the water and has pores filled with air and snow. The keel is the underwater part and can be further separated into an upper completely frozen layer called the consolidated layer, which is always thicker than the surrounding level ice and a lower unconsolidated part that has loose blocks partially refrozen together with water trapped between the blocks (ISO, 2010).

Generally, ice ridges are subdivided into first-year versus older ice ridges. During its first winter and summer, an ice ridge is called a first-year ice ridge (e.g. Figure 3). The consolidation process in the keel part progresses with time and the keel part is closed to fully consolidated if the ridge has survived one summer's melt. Ridges that survive one or more summers are referred to as old ice ridges.

In this work, first-year ice ridges are considered for design of ice-capable vessels. For one thing, the ice conditions along the commercial Arctic shipping routes, such as the Northern Sea Route, are mostly first-year and few ice appears in summer seasons. For another, fewer studies have been made on old ice ridges than the first-year ice ridges and studies for the mechanical and physical properties of the multi-year ice ridges are very limit.

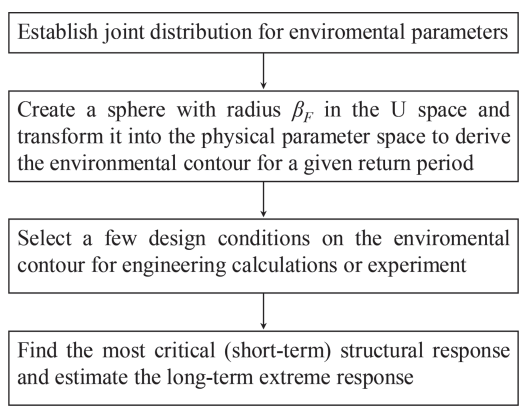

Figure 2. Flowchart of the environmental contour method used for approximate the long-term extreme response.

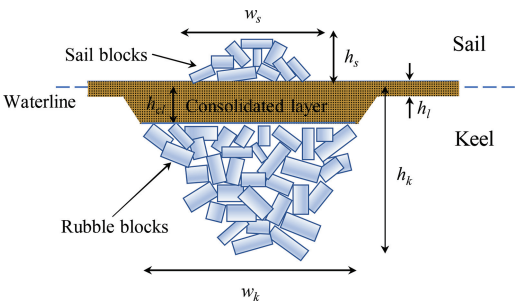

Figure 3. First-year ice ridge with some key parameters: sail draft h_s, sail width ws, consolidated layer thickness h_{cl}, surrounding level ice thickness h_l, keel draft h_k and keel width w_k.

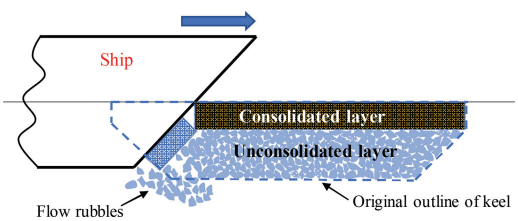

Figure 4. Illustration of ship interacting with a typical first-year ice ridge.

For the scenario of an ice-capable ship interacting with a first-year ice ridge, the ship structure can be simplified as a downward sloping structure. The effects of ridge sail can be neglected since the volume of the sail is small compared to that of the keel part (ISO, 2010). Former studies on a ridge failure against a confederation bridge, whose piers are designed as (slope) ice-breaking cones, have shown that failure of the consolidated layer is the dominant term in determining the keel loads due to the ice ridge interaction with the slope structure. Also, there is no correlation between the keel depth, h_k, and the keel loads. Based on former studies for first-year ice ridge interacting with slope structures, the ship-ice ridge interaction process is illustrated in Figure 4. Flow rubbles from the unconsolidated layer would be cleared by the local water current. The action due to the consolidated layer part can be approximated as level ice with an equal thickness that interacts with the vessel and the mechanical properties of the consolidated layer are assumed to be close to those of level ice. Therefore, the ship-ridge interaction process is further simplified as a ship-level ice interaction event for the purpose of preliminary design.

The ship-level ice interaction process is initiated by a localized crushing of the ice edge and then the contact area between the ship and the ice sheet as well as the crushing force both increase when the ship is advancing and penetrating the ice features. The ice sheet eventually deflects and the bending stresses imply a flexural failure at a certain distance from the crushing region (Jordaan, 2001). Therefore, the ice thickness, the crushing strength and the flexural strength of the (equivalent) level ice are considered as the key parameters for determining the loads due to ship-ice ridge interaction process at the early design stage.

4 PROBABILISTIC MODELS

In this section, probabilistic models are applied in order to describe the distributions of the above-mentioned key parameters which determines the loads due to ship-ice ridge interaction. These key parameter are believed to be dependent on geographical location and season, etc. In this work, the first-year ice ridges in the Barents Sea are considered since many field experiments have been performed in this area by relevant Norwegian and Russian research institutes.

The average thickness of the consolidated layer, based on measurements which are collected by mechanical drilling, can be described by a Gamma distribution (Strub-Klein and Sudom, 2012):

$$f(h_{cl}) = \frac{1}{\Gamma(k)\theta^k} h_{cl}^{k-1} \exp\left(-\frac{h_{cl}}{\theta}\right) \tag{2}$$

where k and θ are the shape and scale parameters for the Gamma distribution, respectively. Based on the experimental data, these two values are determined to be 2.97 and 0.54.

The fitted and sample probability density function of the consolidated layer thickness are plotted in Figure 5. However, sample data for the flexural

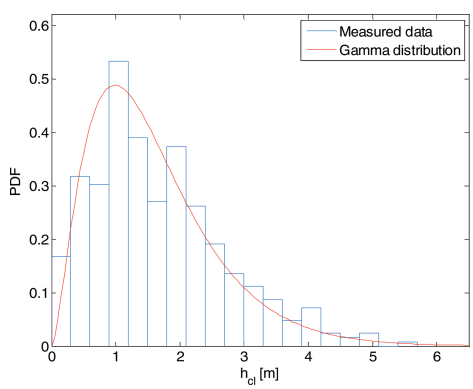

Figure 5. Probability distribution for the consolidated layer thickness h_{cl}.

strength σ_f and the crushing strength σ_c of the consolidated layer is very limited. The mechanical properties of the surrounding level sea ice can serve as effective alternatives (Timco et al., 2000). On the basis of the experimental data for the flexural strength of the level ice in the Barents Sea, the marginal PDF (probability density function) of the flexural strength is described by a two-parameter Weibull distribution (Krupina and Kubyshkin, 2007):

$$f(\sigma_f) = \frac{\beta}{\alpha}\left(\frac{\sigma_f}{\alpha}\right)^{\beta-1}\exp\left(\left(\frac{\sigma_f}{\alpha}\right)^{\beta}\right) \tag{3}$$

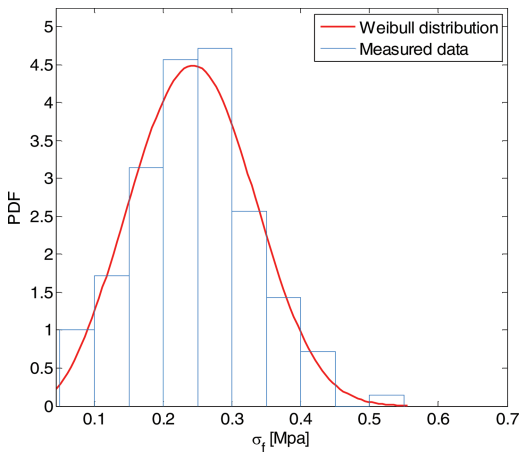

Figure 6. Probability distribution for the flexural strength σ_f.

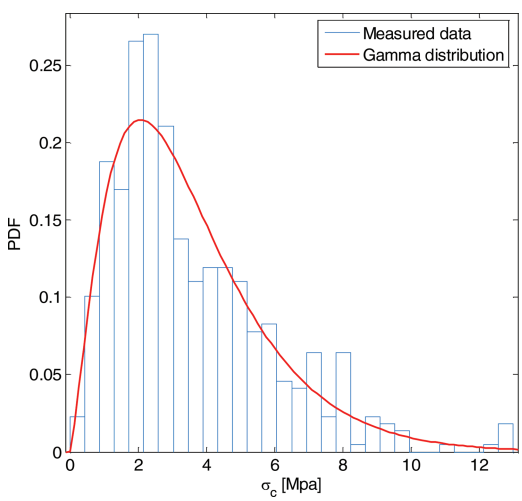

Figure 7. Probability distribution for the crushing strength σ_c.

where the scale parameter $\alpha = 0.274$ and the shape parameter $\beta = 3.167$ are obtained from actual data. The marginal PDF and the empirical histogram of the flexural strength are presented in Figure 6.

Full-scale measurements have been performed to collect the data of the crushing (or compressive) strength of the level ice in the Barents Sea. The distribution of the crushing strength for the vertically loaded samples can be described by the Gamma distribution given in equation (1) with a shape parameter of 2.5 and a scale parameter of 1.40 (Strub-Klein, 2017). The marginal PDF of the of the crushing strength and the corresponding empirical histogram based on experimental data are shown in Figure 7.

5 ENVIROMENTAL CONTOUR

Assume that the desired ice-capable ship mainly sails and operates in the Barents Sea with an annual voyage length of 5000 km in ice ridge fields. The ice ridge density is 2/km along the route. Therefore, for a 50-year return period, the failure probability P_f is determined as:

$$P_f = 1/(50 \cdot 5000 \cdot 2) \tag{4}$$

Correspondingly, the radius of the sphere in the U space, β_F is equal to 4.611 according to equation (1).

Transformation of the sphere in the U space into the environmental contour in the physical parameter space is generally based on the inverse Rosenblatt transformation (DNV, 2010). However, in order to perform such a transformation, the joint distribution of the environmental parameters as described by the conditional modeling approach is required. This necessitates a great amount of sampled data. Due to the limitation of experimental data for the first-year ice ridge statistics, only the marginal PDFs of the abovementioned key parameters can be obtained. Nevertheless, the joint distribution of the environmental parameters with consideration of the correlations between these environmental variables can be approximated by the Nataf distribution model (Silva-González et al., 2013). Then, the environmental contours can be obtained by the Nataf transformation.

Let the variables S_1, S_2 and S_3 represent the consolidated layer thickness, the flexural strength and the crushing strength, respectively. The symbols ρ_{12}, ρ_{13} and ρ_{23} denote the correlation coefficients between these variables. The Nataf transformation model is given as:

$$F_{S_1}(s_1) = \Phi(u_1) \tag{5}$$

$$F_{S_2}(s_2) = \Phi(u_2\sqrt{1-\rho_{12}'^2} + \rho_{12}'\Phi^{-1}(F_{S_1}(s_1))) \tag{6}$$

$$\begin{aligned} F_{S_3}(s_3) = \Phi(&\frac{u_3}{\sqrt{1-\rho_{12}'^2}}\sqrt{1-\rho_{12}'^2-\rho_{13}'^2-\rho_{23}'^2+2\rho_{12}'\rho_{13}'\rho_{23}'} \\ &+\frac{1}{1-\rho_{12}'^2}(\rho_{13}'-\rho_{12}'\rho_{23}')\Phi^{-1}(F_{S_1}(s_1)) \\ &+\frac{1}{1-\rho_{12}'^2}(\rho_{23}'-\rho_{12}'\rho_{13}')\Phi^{-1}(F_{S_2}(s_2))) \end{aligned} \tag{7}$$

where U_1, U_2 and U_3 represent independent standard normal variables and their values can be obtained based on the following equation:

$$\sum_{i=1}^{3} u_i^2 = \beta_F^2 \tag{8}$$

The coefficients ρ_{ij}' $(i, j = 1, 2, 3;\ i \neq j)$ are the corresponding (equivalent) correlation coefficients used in the Nataf transformation and their relationship with ρ_{ij} can be approximated by a semi-empirical equation, which is given as:

$$\rho_{ij}' = \xi \cdot \rho_{ij} \tag{9}$$

where relevant description for determining the function ξ can be found in Ref. (Liu and Der Kiureghian, 1986).

There exist few previous studies on the correlation between these three key parameters for sea ice material. For simplicity, we assume that the correlation coefficients ρ_{ij} $(i, j = 1, 2, 3;\ i \neq j)$ are equal to 0.5. Correspondingly, the 50-year contour surface for the three key parameters is obtained by the Nataf model described by equations (5)–(9) and it is plotted in Figure 8.

The correlation coefficients in this simple case are chosen somewhat arbitrarily due to a imitated amount of information in the literature. Therefore, the influence of the correlation coefficient on the shape of the environmental contour is studied in a parametric way. For simplicity, a two-dimensional contour line on the contour surface is selected for this sensitive study. The consolidated layer is fixed as its mean value 1.59 m, and the corresponding two-dimensional contour lines for different values of the correlation coefficient between the flexural strength and the crushing strength are plotted in Figure 9. In addition, for the sensitive study, ρ_{12} and ρ_{13} are kept at a fixed value equal to 0.5.

It is seen in Figure 9 that the variation of the correlation coefficient has a quite important effect on the shapes of the contour lines. The maximum values of the flexural strength and the crushing strength on the contour line do not change with the correlation coefficient. However, the critical region with large values of the flexural strength and crushing strength becomes narrow as the correlation coefficients increase. Narrower critical region implies that large values of the flexural strength are more easily (or more possibility) to be accompanied by large values of the crushing strength, which would cause serious structural response. Therefore, the correlation coefficients has significant influence on the extreme response for the ice-capable vessel sailing in the ice ridge field.

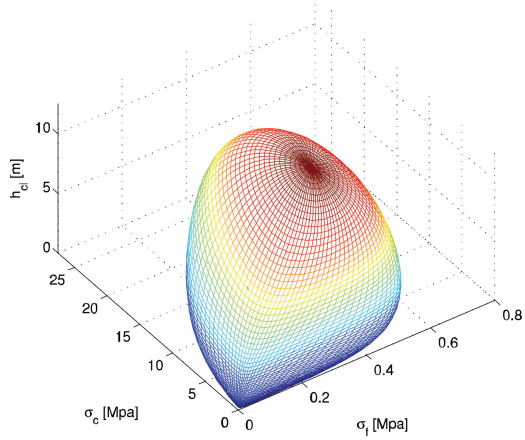

Figure 8. The 50-year contour surface for the consolidated layer thickness, flexural strength and crushing strength.

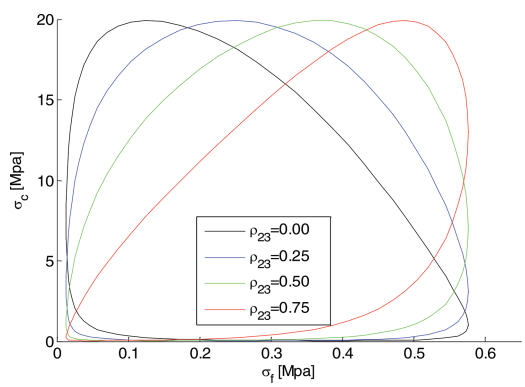

Figure 9. Influence of the correlation coefficient on the two-dimensional contour lines.

6 CONCLUSIONS

In this work, the environmental contour method was proposed for design of ice-capable vessels sailing in areas with ice ridges. Environmental

contours for the key parameters of the first-year ice ridge have been established by applying the Nataf model.

Based on the abovementioned simple case, it is found that variation of the correlation coefficients between the environmental parameters has a very important influence on the shapes of the environment contour as well as on the extreme response of the vessel. Therefore, a reliable numerical model is important in order to investigate this influence.

Furthermore, better knowledge for the correlations between these key parameters, either provided by experiments or numerical simulations, will promote the development of reliability-based design of ice-capable vessels in Arctic regions.

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REFERENCES

Baarholm, G.S., Haver, S. & Økland, O.D. (2010) Combining contours of significant wave height and peak period with platform response distributions for predicting design response. *Marine Structures,* 23, 147–163.

Chai, W., Naess, A., Leira, B.J. & Bulian, G. (2016) Efficient Monte Carlo simulation and Grim effective wave model for predicting the extreme response of a vessel rolling in random head seas. *Ocean Engineering,* 123, 191–203.

DNV (2010) Recommended practice DNV-RP-C205: Environmental conditions and environmental loads. *Norway: Det Norske Veritas.*

Giske, F.-I.G., Leira, B.J. & Øiseth, O. (2017) Full long-term extreme response analysis of marine structures using inverse FORM. *Probabilistic Engineering Mechanics*.

Haver, S. & Winterstein, S.R. (2009) Environmental contour lines: A method for estimating long term extremes by a short term analysis. *Transactions of the Society of Naval Architects and Marine Engineers,* 116, 116–127.

Høyland, K.V. (2014) Ice ridge characteristics and engineering concerns regarding ice ridges. *Proc IAHR Int Symp on Ice.* Singapore.

ISO (2010) 19906: Petroleum and Natural Gas Industries–Arctic offshore structures. *Geneva: ISO.*

Jordaan, I.J. (2001) Mechanics of ice–structure interaction. *Engineering Fracture Mechanics,* 68, 1923–1960.

Krupina, N.A. & Kubyshkin, N.V. (2007) Flexural Strength of Drifting Level First-year Ice in Barents Sea. *International Journal of Offshore and Polar Engineering,* 17.

Leira, B.J. (2008) A comparison of stochastic process models for definition of design contours. *Structural Safety,* 30, 493–505.

Li, Q., Gao, Z. & Moan, T. (2016) Modified environmental contour method for predicting long-term extreme responses of bottom-fixed offshore wind turbines. *Marine Structures,* 48, 15–32.

Liu, P.-L. & Der Kiureghian, A. (1986) Multivariate distribution models with prescribed marginals and covariances. *Probabilistic Engineering Mechanics,* 1, 105–112.

Silva-González, F., Heredia-Zavoni, E. & Montes-Iturrizaga, R. (2013) Development of environmental contours using Nataf distribution model. *Ocean Engineering,* 58, 27–34.

Strub-Klein, L. & Sudom, D. (2012) A comprehensive analysis of the morphology of first-year sea ice ridges. *Cold Regions Science and Technology,* 82, 94–109.

Strub-Klein, L. (2017) A Statistical Analysis of First-Year Level Ice Uniaxial Compressive Strength in the Svalbard Area. *Journal of Offshore Mechanics and Arctic Engineering,* 139, 011503.

Timco, G., Croasdale, K. & Wright, B. (2000) An overview of first-year sea ice ridges. *PERD/CHC report*, 5–112.

Winterstein, S.R., Ude, T.C., Cornell, C.A., Bjerager, P. & Haver, S. (1993) Environmental parameters for extreme response: Inverse FORM with omission factors. *Proceedings of the ICOSSAR-93, Innsbruck, Austria*, 551–557.